

Quantifying the dependent failure of urban power-water supply systems under earthquake hazards from the perspective of energy transfer

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ABSTRACT

Urban power-water systems (PWSs) exhibit complex dependence that exacerbates the degradation of their infrastructure service capacity under earthquake hazards. Existing earthquake risk analysis frameworks for PWSs usually characterize the system dependencies as binary states through joint failure probabilities, which fail to capture the dynamics of cascading failure among dependent PWSs. From the perspective of energy transfer among the dependent infrastructure systems, this study proposes a novel seismic performance analysis framework for urban PWS to reveal the energy transfer amid the power system (PS), water distribution system (WDS), and terminal users, which integrates power flow and water hydraulic models. A dual-network model is first constructed to characterize PWS with directional dependency between power substations and water treatment plants. Seismic damage samples of PWS are then generated by seismic fragility models of components and quasi-Monte Carlo simulations. Finally, two real-world PWS cases are employed to validate the effectiveness of this framework. Results show that the proposed framework is capable of depicting the WDS performance that lies between the fully and non-power supply, which is usually higher than the estimations by the previous binary-dependent failure model of the PWS, providing a new perspective for disaster risk reduction of PWS.

1. Introduction

The normal operation of modern urban infrastructure systems relies on the dependence between distinct systems, which not only expands the functional scope of individual infrastructure systems, but also simultaneously increases the vulnerability [1,2]. For example, the power system (PS) provides energy to the pumping stations within the water distribution systems (WDS). Because of the strong dependencies between the two infrastructure systems, earthquake-induced failure of components within the PS would lead to impaired functionality of the associated water pumps, disrupting the reliability of power-water systems (PWSs) under earthquake hazards.

Over the past several decades, recurrent earthquake hazards worldwide have caused extensive damage to urban infrastructures, which significantly disrupted the normal functioning of urban areas [3]. For instance, earthquake-induced water pipeline leakages and breakages have led to post-earthquake unmet water demand for urban residents, hospitals, and rescue operations [4]. Similarly, the PS exhibits high

vulnerability under earthquake hazards, and there is not only the potential for widespread power outages, but also severe impacts on the WDS performance due to their strong dependencies. Therefore, it is essential to conduct earthquake hazard analysis for PWSs. Such interdependencies pose challenges to the disaster risk reduction of the critical urban infrastructure systems, which are more complex than individual systems [5].

The complexity in performance evaluation of interdependent critical infrastructure systems derives from two types of uncertainty factors: modeling uncertainty (parameter and model output uncertainty) and inherent system uncertainty (physical and operational uncertainty) [6]. On the one hand, the first challenge in investigating the seismic performance of PWS is selecting an appropriate method to characterize their topological and functional parameters. Ouyang [7] has classified methods for evaluating PWS performance into five categories, namely empirical approaches [8], input-output-driven methods [9], agent-based simulations [10], system dynamics-based frameworks [11], and network-based methods [12]. The former four methods exhibit

Abbreviations: ACOPF, Alternating current optimal power flow; IM, Seismic intensity measure; PWSs, Power-water systems; PGV, Peak ground velocity; QMC, Quasi-Monte Carlo; SSIQ, System service index of WDS; WTPs, Water treatment plants; DCOPF, Direct current optimal power flow; MC, Monte Carlo; PS, Power system; PGA, Peak ground acceleration; RR, Repair rate; SSIP, System service index of PS; WDS, Water distribution system.

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limitations in evaluating seismic performance of the PWS under earthquake hazard, including inability to characterize the dependency intensity, challenges in model parameter calibration, and failure to distinguish the damage states of system components. However, network-based approaches (including topology-based and flow-based methods) offer advantages such as more intuitive characterization of PWS topological features (nodes representing water sources, pumping stations, substations, and power plants; edges representing water supply pipelines and transmission lines), and the ability to reflect the seismic intensity experienced by system components, which have been extensively researched and applied in this field. For example, Adachi and Ellingwood [13] proposed a topology-based analysis method for assessments of the seismic performance within the WDS to capture dependencies on the PS. Duenas-Osorio et al [14] developed the topology-based seismic fragility model of the PWS. Anwar et al [15] explored a flow-based analytical model to assess the seismic performance of PWS, with a specific focus on examining the interdependencies between water pumps and electrical substations. Li and Zhang [16] developed a flow-based model for the PWS and analyzed the seismic performance evolution for different system performance metrics. The findings indicate that in comparison with topology-based models, flow-based performance metrics yield more accurate assessments of the seismic performance of PWS.

Alternatively, the analysis of the dependent failure process is the second challenge in the assessment of the seismic performance of PWS. The transmission medium of the PS is electromagnetic waves, while the transmission medium of the WDS is water flow. These two systems exhibit substantial heterogeneity across dimensions, including topological characteristics, transmission properties, operational mechanisms, and response behaviors. How to integrate heterogeneous infrastructure systems with substantial heterogeneity for seismic performance evaluation is an urgent issue requiring more attention. In the research on the seismic performance assessment of PWS, the dependent failure models fall into two categories: binary-dependent failure model (i.e., substation failure results in water pump malfunction) and joint probabilistic dependent failure model (i.e., incorporating the earthquake-induced failure probabilities of substations and pump stations). For example, Ouyang [17] employed the binary-dependent failure model to consider the interdependent influence between the WDS and the PS, and conducted response analysis of PWS under local disturbances. Yang et al [18] utilized the binary-dependent failure model to consider the impact of dependency relationships between WDS and PS on the service functionality of building facilities in the recovery process optimization based on reinforcement learning. Xiao et al [19] employed a joint probabilistic failure model to analyze the seismic resilience of interdependent engineering systems, an approach similar to the one used by Li and Zhang [16]. However, these dependent failure models are essentially binary-dependent failure models that treat the systems as operating independently.

In the propagation of cascading failures within the PWS, substation damage does not always lead to complete water pump failure, which depends on the extent of substation functional loss and the proportion of power that the pumps consume. For example, based on reconnaissance reports from the 1994 Northridge earthquake, Çağnan et al [20] pointed out that damaged substations consist of multiple subsystems, and the failure of some components only leads to a partial functional derating of the substation, rather than a complete shutdown. Furthermore, damage records from the 2008 Wenchuan earthquake demonstrate that the total water supply of the water treatment plants in Deyang City decreased by 17.8%, while Guangyuan City experienced a 37.5% reduction [21]. Consequently, these historical reconnaissance reports clearly demonstrate that the functional state of the PWS components under seismic hazards is a continuous degradation process, which discrete binary models fail to capture [22]. Therefore, the dependent failure models used in the aforementioned studies are all binary-state models, failing to characterize the multi-state dynamics of the nonlinear interaction

between dependent infrastructure systems [23]. Table 1 summarizes the characteristics of the PWS performance analysis framework in some existing studies.

In summary, several research gaps exist in the seismic performance assessment of PWS: (i) The dependence of PWS essentially involves the conversion of electrical energy into hydraulic potential energy. However, existing studies fail to quantify the energy transfer relationship within such systems; (ii) Evaluating seismic performance of PWS by binary-dependent failure model or joint probabilistic dependency failure model cannot capture the multi-state of functionality and cascading failure process; (iii) Seismic performance models for PWS have typically been validated through a single case study, thereby failing to explore how the system topological and functional attributes affect the seismic performance of PWS.

Given the gaps in the aforementioned studies, this study presents a novel framework to investigate the seismic performance of PWS from the perspective of energy transfer. The objectives of this study include: (i) Develop a dependent failure model of PWS to quantify the energy transfer processes in the PWS based on the power flow model and water hydraulic model; (ii) Quantify the power-water energy disturbance under earthquake hazards by the electrical energy supply rate from adjacent substations to water treatment plant and pump station; (iii) Reveal the influence of system topological and functional characteristics on the seismic performance of dependent PWS.

2. Methodology

The comprehensive procedures for the proposed research framework of this study are shown in Fig. 1. Firstly, the functional and topological parameters of the Power-Water Systems (PWS) are modeled, coupled with a regional seismic hazard analysis within the geographic location of the PWS. Subsequently, earthquake-damaged samples for the PWS are generated from seismic fragility models of the components and quasi-Monte Carlo (QMC) simulations. The second part is the functional analysis for the independent WDS and PS based on the power flow model and hydraulic model, and modifying functional parameters of the PWS based on the earthquake-induced damage states and the dependencies of the components. The third part is to analyze dependency parameters, seismic performance, and case-specific topological parameters of the PWS through two real-world cases across different seismic intensity levels.

2.1. Earthquake hazard analysis

Because of the large spatial extent of such infrastructure systems, uniform intensity measures (*IMs*) without accounting for the location of seismic fault zones will result in an underestimation of the performance of these infrastructure systems [24–26]. This study employs ground motion attenuation models that are from Ref [27]. to acquire peak *IMs* at individual sites for a specified earthquake scenario. For a given earthquake scenario, the *IMs* (i.e., *PGV*, *PGA*) at site *i* can be expressed as:

$$\ln IM_i = \ln \overline{IM}_i + \ln X_i \quad (1)$$

where $\overline{IM}_i$ denotes the median *IM* (cm/s) derived from the ground motion attenuation model. X_i represents the statistical residual value accounting for the uncertainty of IM_i .

2.2. Seismic fragility of components in PWS

This study focuses on evaluating the seismic failure of the PWS at the urban scale. Water distribution system (WDS) is a complex system consisting of a water source (water treatment plant), a water transmission network, and a water distribution network. Consequently, this study accounts for the damage to water distribution pipelines and the impact of power supply degradation on WTPs, which is a widely adopted

Table 1

Comparison of performance analysis frameworks for PWS.

Reference	Assessment method	Dependent relationship	Dependent failure model	Uncertainty
Adachi and Ellingwood [13]	topology-based	$PS \rightarrow WDS$	binary-state	Component failure; Backup power supply
Dueñas-Osorio et al [14]	topology-based	$PS \rightarrow WDS$; $WDS \rightarrow PS$	probabilistic binary-state	Component failure; Earthquake magnitude
Anwar [15]	flow-based	$PS \rightarrow WDS$; $WDS \rightarrow PS$	binary-state	Component failure; Earthquake magnitude
Li and Zhang [16]	flow-based	$PS \rightarrow WDS$; $WDS \rightarrow PS$	probabilistic binary-state	Component failure; Earthquake magnitude
Ouyang [17]	topology-based	$PS \rightarrow WDS$; $WDS \rightarrow PS$	binary-state	Component failure;
Yang et al [18]	topology-based	$PS \rightarrow WDS$; $WDS \rightarrow PS$	binary-state	Component failure; Earthquake magnitude
<i>This study</i>	flow-based	$PS \rightarrow WDS$	<i>continuous-state</i> ; <i>energy-transfer</i>	<i>Component failure</i> ; <i>Earthquake magnitude</i> ; <i>Backup power supply</i>

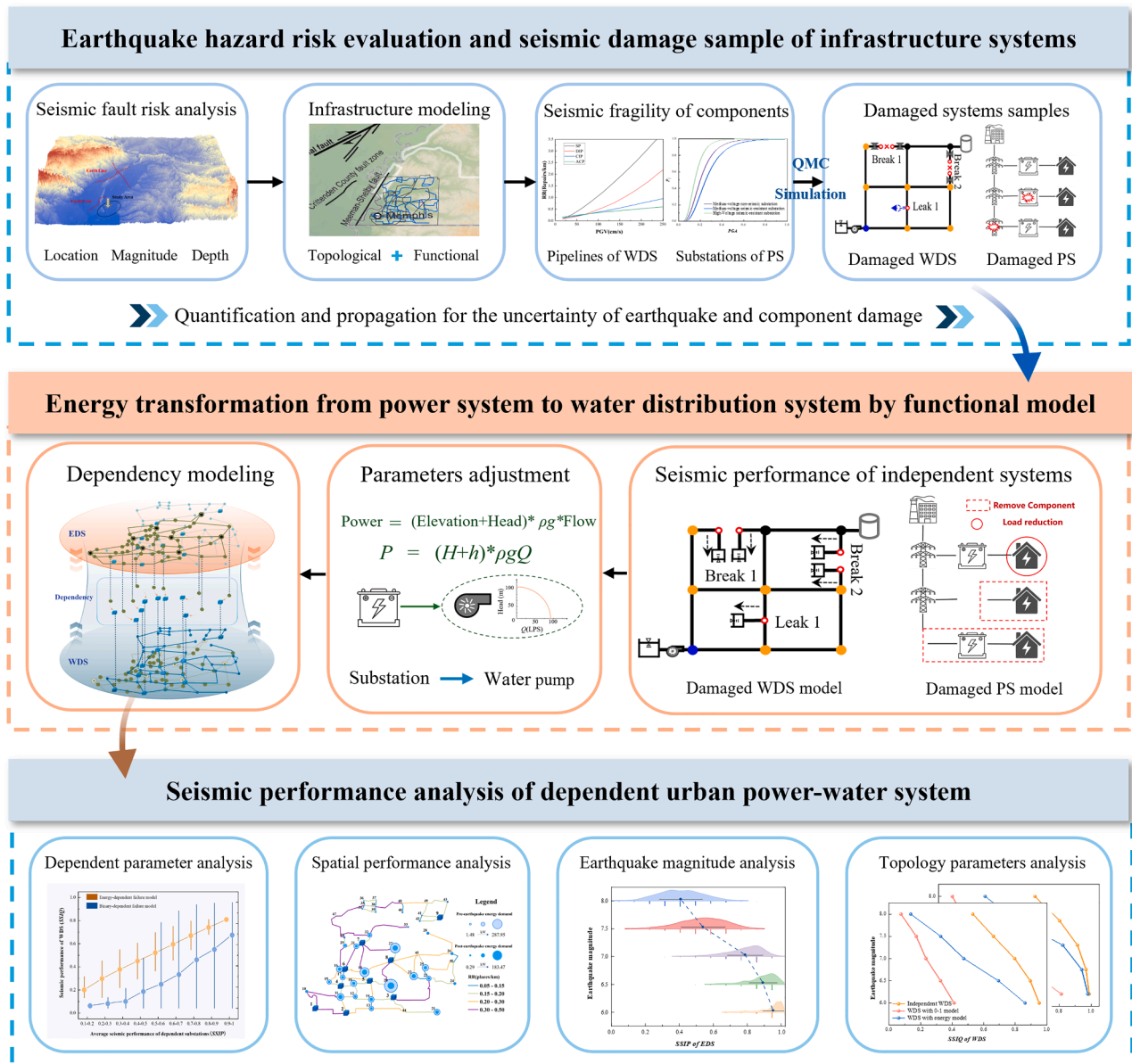


Fig. 1. Framework for the dependent failure analysis of the PWS under earthquake hazards.
(PWS, power-water system; WDS, water distribution system; PS, power system; QMC, quasi-Monte Carlo).

approach in existing seismic risk studies of PWSs [28,31]. Similarly, the power system (PS) analysis is restricted to the distribution network. Therefore, the framework of this study focuses on the damages of power generation plants and substations as the main cause of power loss in the distribution network, which is also a common assumption under disaster scenarios [16,18].

2.2.1. Seismic fragility for pipelines of the WDS

The failure probability of WDS components can be evaluated by seismic fragility models. According to Ref [28], the earthquake-induced pipeline damage usually distributes over the whole urban area and takes the majority of damage in WDS. Earthquake-induced damage to the WDS pipelines relies on the *IMs*, pipeline structural types, and soil characteristics around pipelines, which result in uncertainty in earthquake-induced pipeline damage. American Lifeline Association (ALA) [29] assumes that the number of earthquake-induced damages to pipelines follows a Poisson distribution. The probabilistic parameters of the seismic damage event of the pipeline are estimated using a statistical method based on the pipeline damage data that occurred in historical earthquakes. Subsequently, the damage probability of the pipeline under earthquake hazards can be expressed as follows:

$$P_f = 1 - e^{-RR \cdot L} \quad (2)$$

where *RR* denotes the pipeline repair rate (expressed in repair places per kilometer) associated with earthquake-induced damage, and *L* stands for the total length of the pipeline (in kilometers).

The statistical function was proposed to characterize the relationship between *RR* and the peak ground velocity (*PGV*), with the function derived from historical damage data [28].

$$\begin{cases} RR_0 = A \cdot (PGV - B)^C \\ RR = C_p \cdot C_d \cdot C_g \cdot C_l \cdot RR_0 \end{cases} \quad (3)$$

where *RR*₀ denotes the standard repair rate (expressed in repair places per kilometer), while *C_p*, *C_d*, *C_g*, and *C_l* represent correction coefficients accounting for the impacts of pipeline material, pipeline diameter, topographic geology, and site liquefaction, respectively. *A*, *B*, and *C* are parameters for the standard repair rate function, with specific values referenced in Isoyama et al [30].

2.2.2. Seismic fragility for components of the PS

The damage probability for components of the PS can be computed by seismic fragility models of power generation plants and substations, which assume the damage probability of components obeys a log-normal distribution and the damage occurs independently and randomly [12]. The damage probability for components of the PS is quantified by the equation below:

$$P^{\text{failure}}[D_s \geq D_{si} | PGA] = \phi \left[\frac{1}{\beta_i} \ln \left(\frac{PGA}{\lambda_i} \right) \right] \quad (4)$$

where $P^{\text{failure}}[\bullet]$ represents the probability of exceeding a particular structural damage state under a specified seismic intensity; *D_{si}* represents the structural damage mode; $\phi[\bullet]$ signifies the function of the standard normal distribution; *PGA* refers to the peak ground acceleration; λ_i and β_i denote the median value and logarithmic standard deviation of the fragility model associated with *D_{si}*, respectively.

2.3. Seismic performance analysis of individual system

2.3.1. Seismic performance analysis for the WDS based hydraulic model

Earthquake-induced pipeline damages are classified as leakage or breakage [32]. Based on the HAZUS [33], the proportion of these damage types varies significantly between liquefied and non-liquefied regions: non-liquefied areas see 80% of pipeline damage in the form of leakage and 20% as breakage, while the inverse holds for liquefied

zones, with breakage accounting for 80% and leakage for 20%. The leakage modes are categorized into five types by Shi and O'Rourke [34], which also include occurrence probabilities for each leakage type. According to Refs [3,35,36], the models of leakage and breakage are shown in Fig. 2.

2.3.2. Seismic performance analysis for the PS based power flow

For evaluating the seismic performance of the PSs under earthquake hazards, the optimal power flow (OPF) model is widely adopted in existing research [37,38], with the alternating current optimal power flow (ACOPF) model outperforming the direct current optimal power flow (DCOPF) model in terms of simulating load curtailment processes and calculating power losses with higher precision [39]. Therefore, the ACOPF model is employed to conduct seismic performance of PSs in this study. Eqs. (5) and (6) present the calculation principles for active power and reactive power; Eq. (7) describes the active power demand allocation for the *i* th dependent substation, which is composed of other user loads and water supply loads; and Eq. (8) denotes the ratio of water supply-related load to the overall load of the substation.

$$P_q = \sum_{m=1}^N |V_q| |V_m| (G_{qm} \cos \theta_{qm} + B_{qm} \sin \theta_{qm}) \quad (5)$$

$$Q_q = \sum_{m=1}^N |V_q| |V_m| (G_{qm} \sin \theta_{qm} - B_{qm} \cos \theta_{qm}) \quad (6)$$

$$P_{i,\text{dep}} = P_{i,\text{others}} + P_{i,\text{water}} \quad (7)$$

$$\alpha_{i,\text{water}} = \frac{P_{i,\text{water}}}{P_{i,\text{dep}}} \quad (8)$$

where *P_q* and *Q_q* denote the values of the active power and reactive power corresponding to node *q*, respectively; $|V_q|$ and $|V_m|$ represent the voltage magnitudes corresponding to nodes *q* and node *m*, respectively; θ_{qm} is the voltage phase variance between node *q* and node *m*; *G_{qm}* and *B_{qm}* stand for the real and imaginary parts of the admittance matrix *Y*, respectively; *P_{i,dep}* denotes the rated active power of the substation node that has a dependent relationship with the WDS; *P_{i,others}* is the active power supplied by this dependent substation to other urban users, such as residential and commercial users; *P_{i,water}* is the active power supplied by this substation to the dependent water treatment plant (under normal operation); $\alpha_{i,\text{water}}$ denotes the ratio of the active power supplied by this substation to its dependent water treatment plant.

Under earthquake hazards, components of PSs typically experience direct damage, which further causes the failure of distribution lines connected to the affected substations and disrupts power allocation for other distribution lines; if the power in a distribution line exceeds its capacity, additional line failures will occur. When the power flow in a distribution line exceeds its capacity limit for a sustained period, the overcurrent relay activates to protect the system, causing the line to trip. Therefore, the capacity limit *Cl_i* of distribution line *i* is assumed to be proportional to its initial power flow *L_i⁰* prior to the earthquake, expressed as:

$$Cl_i = (1 + \gamma_i) \cdot L_i^0 \quad (9)$$

where γ_i represents the tolerance parameter for line *i*. When the actual power flow exceeds this capacity, the distribution line will be disconnected.

Fig. 3 presents the overall workflow for assessing the seismic performance of PSs. First, the failure probabilities of components are calculated based on their seismic fragility models and *IMs*. Next, earthquake-induced damage samples of the PS are generated by the MCs, and the power flow model is adjusted according to the damage states of components. Subsequently, island analysis is conducted on the ACOPF model of each damage sample: if there are no generators in an

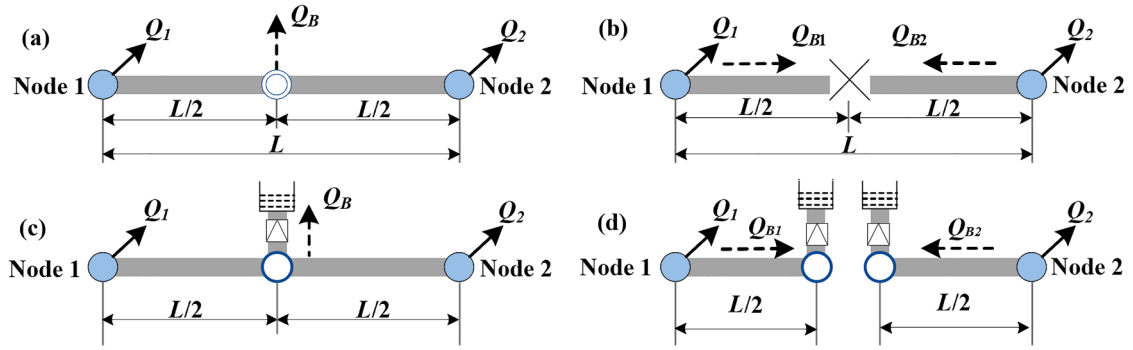


Fig. 2. Pipeline damage patterns and corresponding hydraulic models. (a) Schematic of leakage; (b) Schematic of breakage; (c) Hydraulic model for leakage; (d) Hydraulic model for breakage.

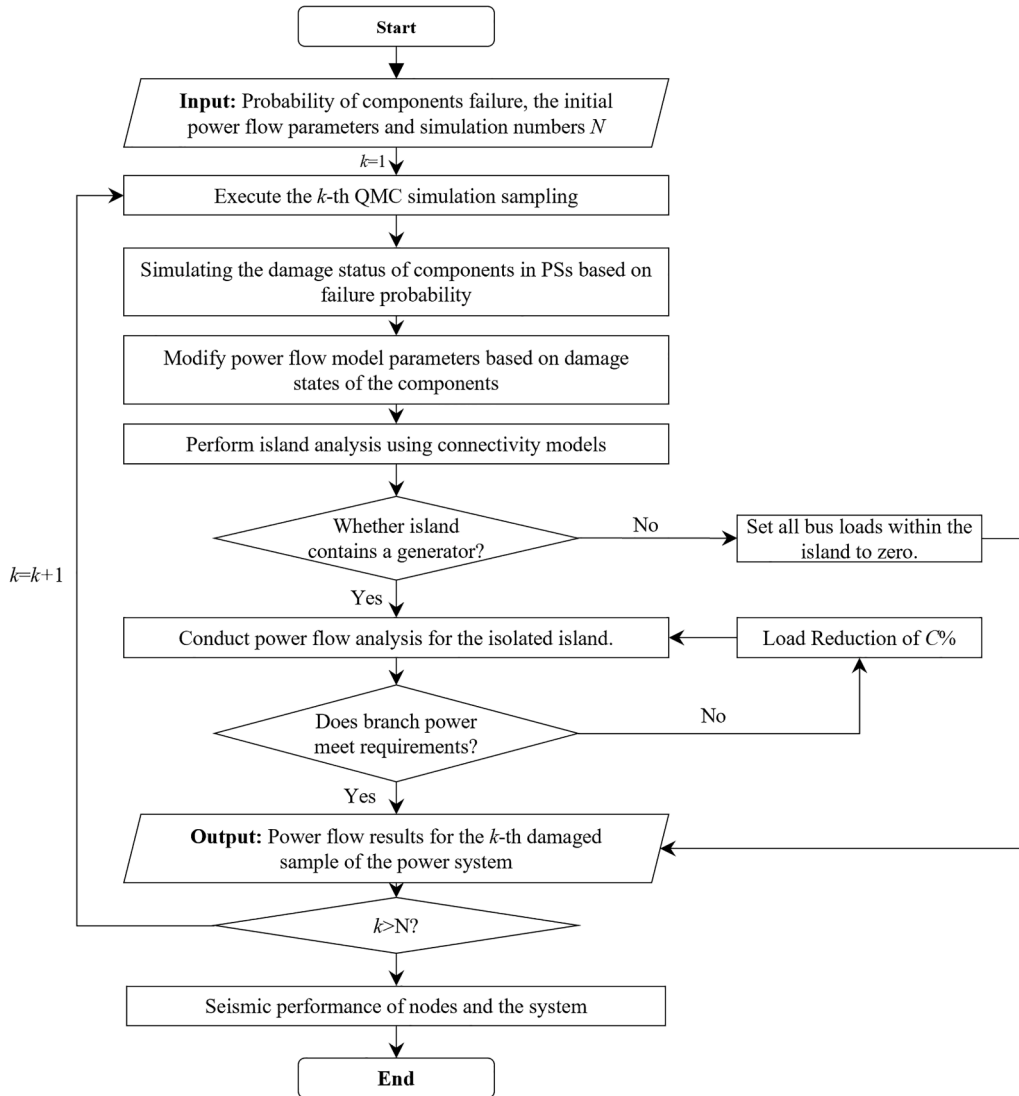


Fig. 3. Flowchart for ACOF-based seismic performance assessment of PS.

island, all loads within it are deemed completely non-functional; if generators are present in an island, the load bus with the largest power in the island is selected as the balance bus to ensure the total generator power matches the total load power, followed by checking whether the load rate of each branch exceeds the threshold and adjusting load power until the load rate of all lines falls below their rated capacity. Finally,

once the power of all load buses and lines meets the constraints, the performance indicators for individual load buses and the overall power system are output.

2.4. Seismic performance analysis for the PWS based on the energy failure model

This study focuses on the short-term seismic performance assessment of PWS specifically at the urban scale. Consequently, this study strictly concentrates on the unidirectional power-to-water dependency. Under earthquake hazards, PSs not only experience power reduction but also cause functional degradation of associated WDS components, potentially resulting in localized or widespread water supply shortages for end-users. Based on reconnaissance reports from the Los Angeles earthquake [20] and Wenchuan earthquake [21], it can be seen that the functional state of the PWS under seismic hazards is better represented by a continuous model than a binary model, which is shown in Fig. 4.

To describe this continuous mapping state of the PWS after an earthquake, this study proposes an energy-dependent failure model. Eq. (10) denotes the actual active power that the i th substation can supply to its dependent water treatment plant after the earthquake, which incorporates parameters of post-earthquake power supply at the substation and the backup power supply to the water treatment plant. Eq. (11) further describes the relationship between backup power availability to the water treatment plant. Eq. (12) represents the ratio of post- to pre-earthquake power of the i th substation. Eq. (13) describes the energy transfer between the power and water pressure by the pumps in the water treatment plant. In this equation, the right side of the equation represents the concept of energy in units of WTP. ρ is water density (kg/m^3), g is gravitational acceleration (m/s^2), Q_e is the volumetric flow rate (m^3/s), H^* is the total nodal head (the sum of elevation head and pressure head, in m), and η_e is the dimensionless total energy conversion efficiency. After dimensional analysis, the fundamental SI (International System of Units) dimension of the right side is identically $\text{kg}\cdot\text{m}^2/\text{s}^3$. The left side of the equation represents the active electrical power consumed by the water pump, measured in Watts (W), with the fundamental SI dimension of $\text{kg}\cdot\text{m}^2/\text{s}^3$.

$$P_{i,\text{water}}^{\text{post}} = P_{i,\text{water}}^{\text{pre}} \cdot \left[\prod (\beta_i \geq \alpha_{i,\text{water}}) + f(S) \cdot \prod (\beta_i < \alpha_{i,\text{water}}) \right] \quad (10)$$

$$f(S) = \prod (S=1) + \beta_i \cdot \prod (S=0) \quad (11)$$

$$\beta_i = P_i^{\text{post}} / P_i^{\text{pre}} \quad (12)$$

$$P_{i,\text{water}}^{\text{post}} = \frac{\rho g}{\eta_e} Q_e(t) H^*(t) \quad (13)$$

where $\prod(\cdot) = \{0, 1\}$ denotes a bivariate indicator function; $P_{i,\text{water}}^{\text{post}}$ is the active power supplied by the i th dependent substation to the water treatment plant and pumps under earthquake hazards; $P_{i,\text{water}}^{\text{pre}}$ is the

active power supplied by the i th dependent substation to the water treatment plant and pumps under normal operation; P_i^{pre} denotes the total active power of the i th dependent substation under normal operation; P_i^{post} denotes the post-earthquake active power of the i th dependent substation; β_i is the post-earthquake functionality ratio of the i th dependent substation, where $\beta_i \in (0, 1)$; $\alpha_{i,\text{water}}$ denotes the ratio of the active power supplied by the i th dependent substation to its dependent water treatment plant and pumps, where $\alpha_{i,\text{water}} \in (0, 1)$; $f(S)$ is the backup power function; S is the backup power supply variable; η_e denotes the power conversion efficiency coefficient of the pumps. $Q_e(t)$ denotes the normal water supply flow of the i th dependent water treatment plant (L/s); $H^*(t)$ refers to the head of the i th dependent water treatment plant (m); ρ is the water density; g is the gravitational acceleration.

In the situation of earthquake hazard, the power supply at the i th dependent substation will be prioritized for the water treatment plant. As shown in Eq. (10), if $\beta_i \geq \alpha_{i,\text{water}}$, the power requirement at the water treatment plant is thus satisfied. Otherwise, if $\beta_i < \alpha_{i,\text{water}}$, it is necessary to determine whether the water treatment plant has a backup power supply. As shown in Eq. (11), if a backup power supply is available ($S = 1$), the water treatment plant will operate as expected follows a probability of $P(W|E)$; if no backup power supply exists ($S = 0$), there won't have the power supply to the water treatment plant.

The assessment workflow for PWS seismic performance based on the energy-dependent failure model is presented in Fig. 5. First, seismic scenarios and the fragility models of PWS components are input in accordance with Sections 2.1 and 2.2. Further, quasi-Monte Carlo (QMC) simulations are executed to produce damage samples for the WDS and PS separately. Then, the functional analysis results of the PS are input into the WDS functional model, and the functional parameters of the WDS are adjusted based on the energy-dependent failure model described above. Finally, the seismic performance analysis results of the PWS are output.

The system service index based on the water demand (SSIQ) for WDS is widely utilized as a quantitative measure to assess and enhance the seismic performance of such systems [31,40,41]. Given the inherent uncertainty associated with earthquake-induced pipeline damage, this study employs QMC simulation to assess WDS seismic performance, with the system-level performance estimated using the average of SSIIQ values across all MC samples.

$$SSIIQ = \frac{1}{N} \sum_{k=1}^N \frac{\sum_{i=1}^m Q_i^k}{\sum_{i=1}^m Q_{0i}} \quad (14)$$

where m denotes the total count of user nodes within the WDS; Q_i^k represents the actual water delivery volume at the i th node in the k -th

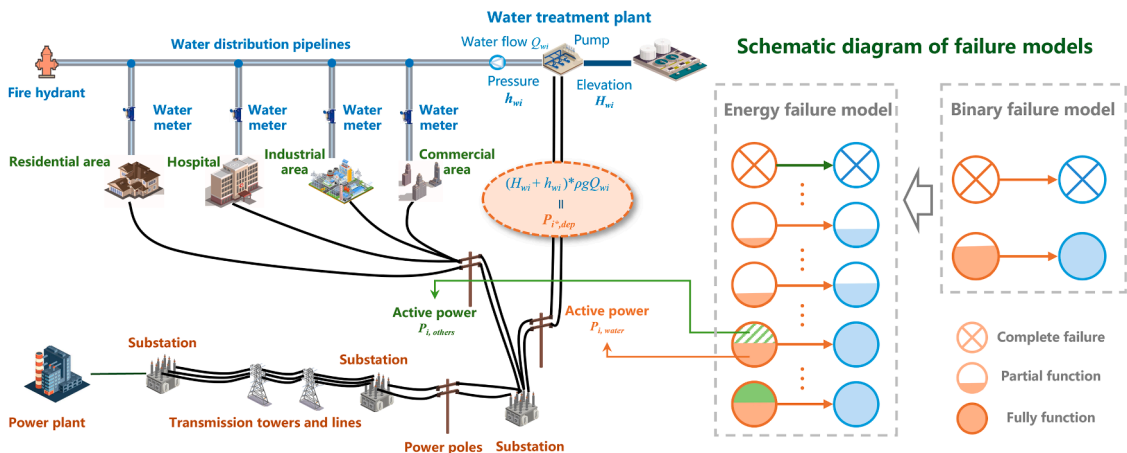


Fig. 4. Schematic of energy transfer process in the PWS.

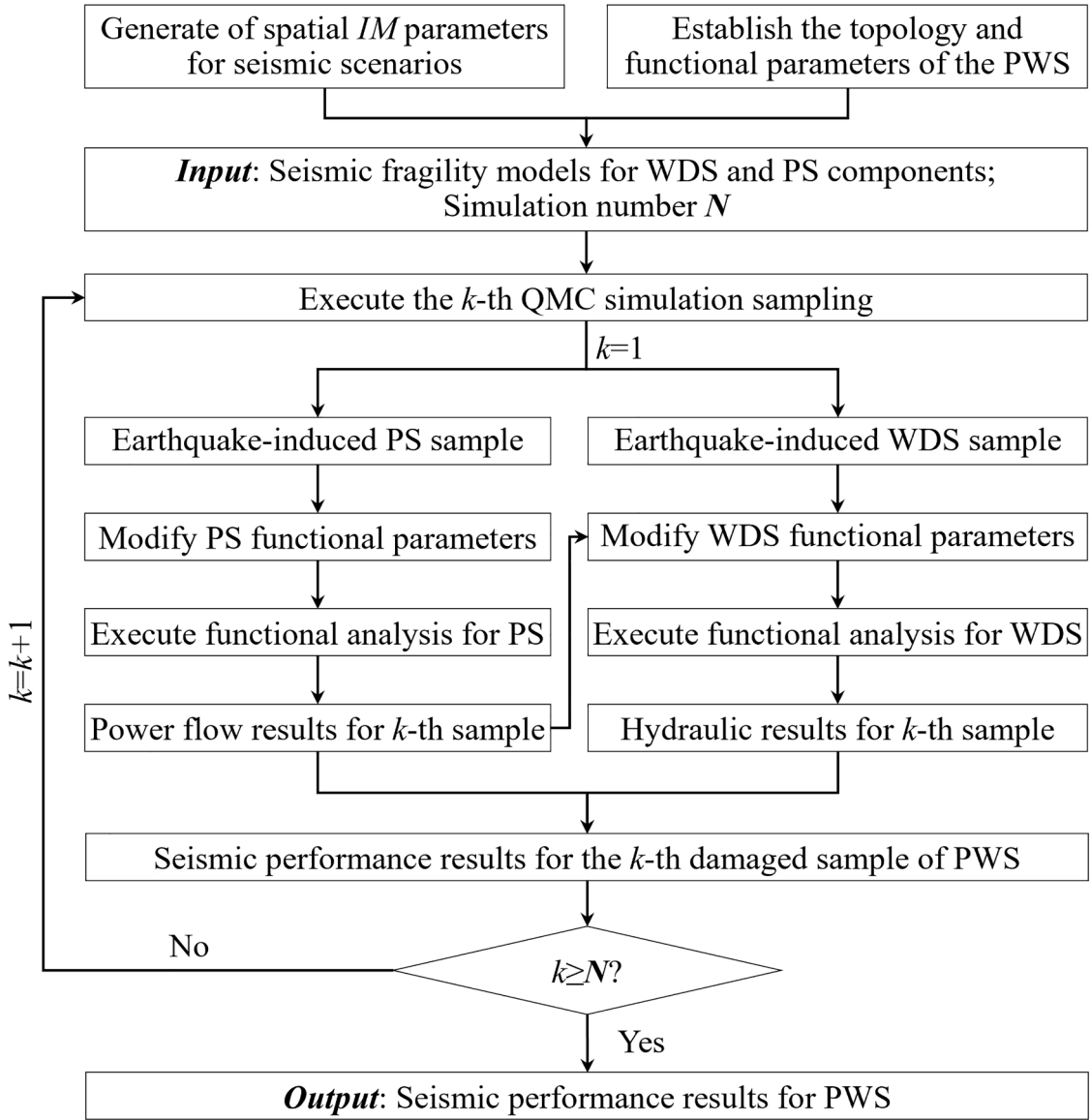


Fig. 5. Flowchart of the seismic performance assessment of PWS based energy-dependent model.

QMC sample; and N stands for the total number of QMC samples employed in the analysis.

Correspondingly, this study adopts the active power satisfaction rate as a measure indicator for assessing seismic performance of the PS, where the system service index of PS ($SSIP$) is determined by averaging the post-earthquake active power values from all QMC samples.

$$SSIP = \frac{1}{N} \sum_{k=1}^N \frac{\sum_{i=1}^n P_i^k}{\sum_{i=1}^n P_{0i}} \quad (15)$$

where n denotes the total count of user nodes within the PS; P_i^k represents the actual active power at i th substation in the k -th QMC sample; and N stands for the total number of QMC samples employed in the analysis.

3. Case study

This section first introduces the power-water system (PWS) case in Shelby County, Tennessee. Subsequently, the analysis of different dependent failure models and the evolution of seismic performance under varying earthquake intensities are performed to verify the

effectiveness of the energy-dependent failure model. Furthermore, the evolution comparison between the Shelby case and the NY case is carried out to reveal how the topological and functional characteristics of PWS influence the seismic performance. Finally, a dependent parameter sensitivity analysis on the seismic performance of PWS is performed to identify critical links and parameters that affect cascading failures in PWS.

The case study of the PWS in Shelby is employed to analyze the seismic performance of PWS using different dependent failure models [42]. As shown in Fig. 6, the water distribution system (WDS) of this PWS comprises 9 water sources, 71 pipelines, and 49 user nodes, whereas its power system (PS) consists of 8 power plants, 21 12-kV substations, 30 23-kV substations, and 71 distribution lines.

This study selects the New Madrid Seismic Zone (NMSZ), situated northwest of Shelby County, with an earthquake magnitude of $M_W=7.5$ and a hypocenter depth of 10 km [43]. Utilizing Eq. (1) provided in Refs [25,26], the peak ground velocity (PGV) and peak ground acceleration (PGA) can be derived, which are shown in Fig. 7. For the substations within the power system (PS), this study designates the “Moderate Damage” state from fragility models as the failure criterion. The failure probabilities of substations are calculated, while those of pipelines are

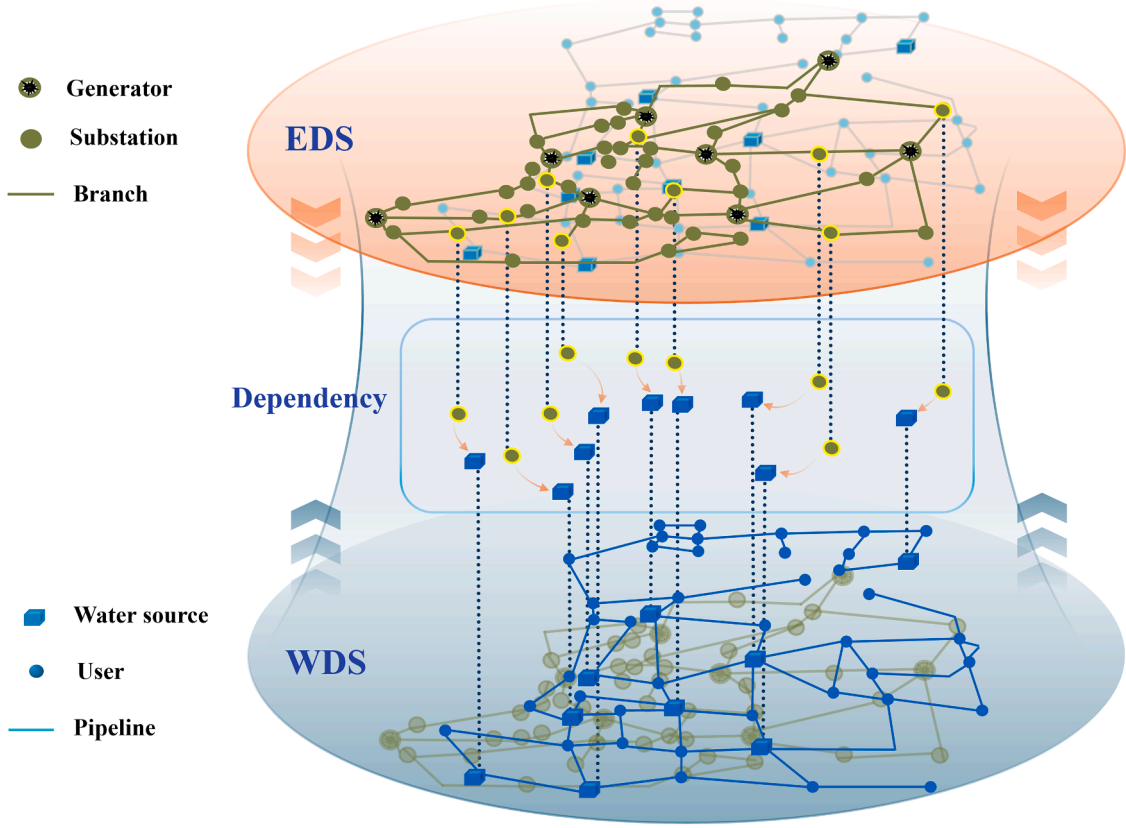


Fig. 6. Diagram of the PWS topology and dependencies for the Shelby.

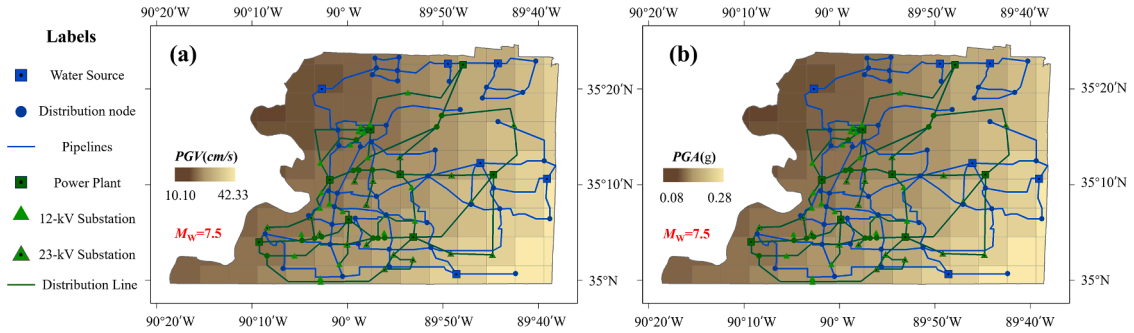


Fig. 7. Earthquake scenario in the case of Shelby ($M_w=7.5$). (a) Spatial distribution of PGV; (a) Spatial distribution of PGA.

computed using Eq. (2).

The seismic performance of PWSs is typically analyzed using Monte Carlo (MC) simulations. To improve computational efficiency and accelerate the convergence of seismic performance results, this study employs quasi-Monte Carlo (QMC) sampling to analyze the seismic performance, which is based on the energy-dependent failure model and the binary-dependent failure model, respectively [44]. Fig. 8(a) presents the findings of the convergence analysis, demonstrating that following 2000 QMC simulations, the seismic performance of the WDS converges under both types of dependent failure models. Compared to the binary-dependent failure model, the energy-dependent failure model achieves faster convergence. Similarly, Fig. 8(b) indicates that the seismic performance errors of the energy-dependent failure model and the binary-dependent failure model are 0.0012 and 0.0033, respectively. Therefore, the count of QMC simulations (N) is configured to 2000.

3.1. Comparative analysis of dependent failure models

The binary-dependent failure model is a dependent failure model commonly adopted in existing research for analyzing the seismic performance of PWS, which has a flaw of ambiguous failure boundary parameters and lacks a specific analysis of failure thresholds. For example, what is the minimum seismic performance threshold of the substation that the water treatment plant depends on to cause the water supply capacity to fail? To address this gap, this section analyzes the seismic performance differences of the PWS in Shelby County under two seismic scenarios with moment magnitudes $M_w=6.0$ and $M_w=7.5$, by adopting various thresholds for the binary-dependent failure model. It should be noted that when the threshold for the binary-dependent failure model is set to 50% (hereafter referred to as Logic-50%), the water treatment plant fails immediately if the post-earthquake power output of the substation is $<50\%$ of its pre-earthquake power output ($\beta_i < 50\%$).

As shown in Figs. 9(a) and 9(b), all of the average service indicators of the substations (SIP) exceed 0.85 when the $M_w=6.0$. Meanwhile, in

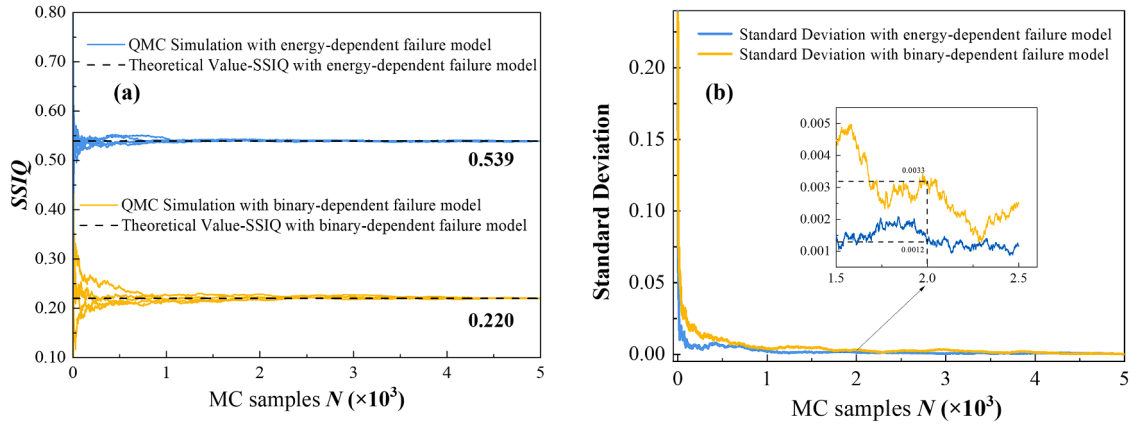


Fig. 8. Convergence analysis of seismic performance for PWS.
(a) Convergence analysis of SSIQ values. (b) The standard deviation for the simulation.

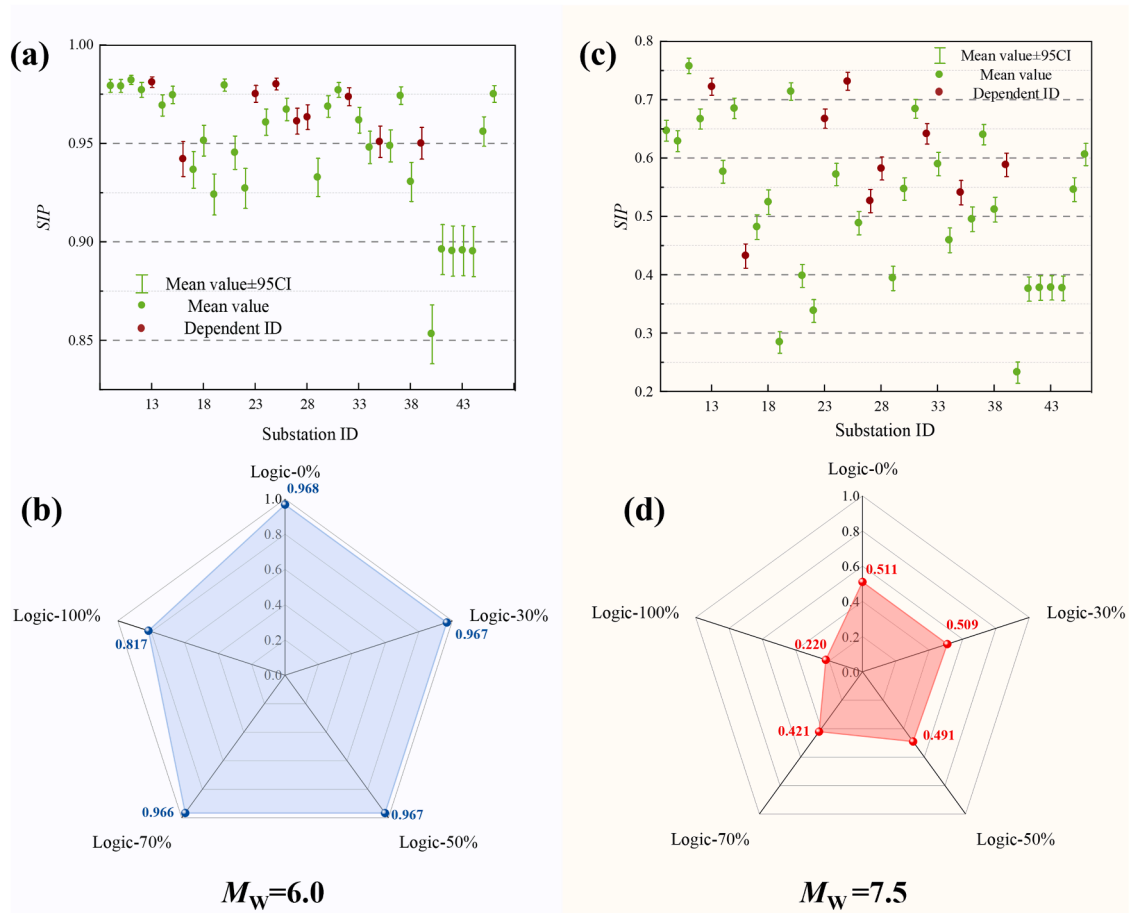


Fig. 9. Seismic performance of PWS under different thresholds of binary-dependent failure models. (a) Seismic performance for the substations of PS (SIP) at $M_W=6.0$. (b) Seismic performance of WDS (SSIQ) at $M_W=6.0$. (c) Seismic performance for the substations of PS (SIP) at $M_W=7.5$. (d) Seismic performance of WDS (SSIQ) at $M_W=7.5$.

the binary-dependent failure model, the results for assessing seismic performance of WDS at the 100% threshold differ from those at other thresholds by around 18.4%. As shown in Figs. 9(c) and 9(d), the average SIP of the substations ranges from 0.2 to 0.8 when the $M_W=7.5$, which leads to significant differences in the seismic performance of WDS across different thresholds. With the increase of the threshold value, the SSIQ of the WDS gradually decreases, and the variation between its seismic performance at the 100% threshold and those at other threshold

levels is 57.3% to 90.9%. This indicates that the binary-dependent failure model is overly reliant on its threshold parameters: as the complexity of failure scenarios increases, selecting different threshold parameters amplifies discrepancies in assessment results.

To analyze the differences in water treatment plant functionality under discrete power supply states (0–100%), this section continues to take the WDS in Shelby County under $M_W=7.5$ earthquake as the analysis case. It should be noted that the pipelines of the WDS did not

sustain direct damage due to seismic effects, and the performance decline was solely caused by the seismic performance degradation of the nine dependent substations in this section. This section compares the results from the binary-dependent failure model (with thresholds set at 70% and 100%) with those from the energy-dependent failure model, which is shown in Fig. 10 (the points represent the mean value, and the lines represent the range of the results). As shown in Figs. 10(a) and 10(b), the horizontal axis denotes the range of average seismic performance of the nine dependent substations, while the vertical axis represents the overall SSIQ of the WDS. When the substation threshold is set to 70%, the average SSIQ curve under the binary-dependent failure model exhibits a distinct stepwise trend with the increase in the average SIP of the dependent substations. At the same average substation performance level, the seismic performance range of the WDS obtained from the binary-dependent failure model is more dispersed than that from the energy-dependent failure model. For substation performance exceeding the threshold, the SSIQ of the WDS assessed by the binary-dependent failure model is greater than that obtained through the energy-dependent failure model, with the remaining outcomes being lower. When the threshold is set to 100%, the SSIQ of the WDS evaluated by the binary-dependent failure model is lower than that by the energy-dependent failure model, regardless of the seismic performance of substations. This indicates that evaluating the impact of power systems on WDSs using the energy-dependent failure model eliminates the influence of threshold selection, which provides a more uniform, concentrated, and continuous characterization of how varying power supply levels affect water treatment plant functionality.

3.2. Seismic performance analysis for the PWS

To analyze the differences in PWS performance between various dependent failure models and the completely independent PWS model, this section conducts a comparative analysis of seismic performance for the Shelby case under different seismic intensities within the same seismic fault zone (binary-dependent failure model with a threshold of 100%). This section selects five seismic scenarios ($M_W = 6.0$, $M_W = 6.5$, $M_W = 7.0$, $M_W = 7.5$, and $M_W = 8.0$), and the analysis results have been presented in Fig. 11. As presented in Fig. 11(a), the average SSIP of the PS gradually decreases from 0.945 to 0.405 across earthquake magnitudes ranging from $M_W = 6.0$ to $M_W = 8.0$. As shown in Fig. 11(b), compared to the independent WDS, the average SSIQ of the WDS is lower when considering the PS dependencies, regardless of the type of dependent failure model employed. This highlights the necessity of accounting for the impact of the PS dependencies when assessing the SSIQ of WDS.

The SSIQ of the WDS calculated by the energy-dependent failure

model presented in this study falls between that of the independent operation model and the binary-dependent failure model across all earthquake magnitudes. This indicates that the binary-dependent failure model typically overestimates the seismic performance loss of the WDS. However, as earthquake intensity increases, the variation between the energy-dependent failure model and the independent operation model in assessing the WDS gradually increases from 5.17% to 60.39%, while the variation between the energy-dependent failure model and the binary-dependent failure model diminishes from 94.8% to 39.61%. This occurrence is attributable to the extensive pipeline damage that leaves the WDS almost paralyzed during extremely rare earthquakes, meaning the choice of dependent failure models for the PWS exerts minimal influence on the seismic performance outcomes of the WDS.

To further analyze the energy supply relationship between substations and water treatment plants, the $M_W = 7.5$ seismic scenario is used as an example, and the probability density distribution of damaged samples results is presented in Fig. 12. A comparison of Figs. 12(a) and 12(b) shows that the energy demand of water treatment plants ranges from 50 kW to 600 kW, while the active power supplied by substations ranges from 18 MW (18 000kW) to 120 MW (120 000kW). This indicates that the power demand of water treatment plants in the WDS accounts for a small proportion of the total load of substations. However, a comparison of Figs. 12(a) and 12(c) reveals that due to the islanding effect between substations (i.e., the failure of adjacent substations can lead to regional blackouts), the water supply of water treatment plants in the WDS is highly prone to complete failure after an earthquake. Consequently, the vulnerability of the PS under the earthquake disaster can significantly affect the functionality of water treatment plants, thereby impacting the entire water supply within the city.

Fig. 13 also spatially displays the energy values (equivalent power) of each user node. From Fig. 13(b), it can be observed that the three nodes of the WDS with the highest pre-earthquake energy values (Nodes 19, 20, and 27) exhibit relatively low post-earthquake energy satisfaction rates despite being directly connected to water sources. Conversely, nodes with lower pre-earthquake energy demand (e.g., Nodes 24, 26, and 28) show relatively higher post-earthquake energy satisfaction rates. This phenomenon may be attributed to the vulnerability of PS after the earthquake (as shown in Fig. 13(a)): the PS fails to supply sufficient power to the connected water treatment plants to support the normal operation of their pumps. However, the available power remains adequate for users with low energy demand, resulting in higher post-earthquake energy satisfaction rates for these users.

3.3. Influence of PWS characteristics on seismic performance

To investigate the seismic performance differences among PWS cases

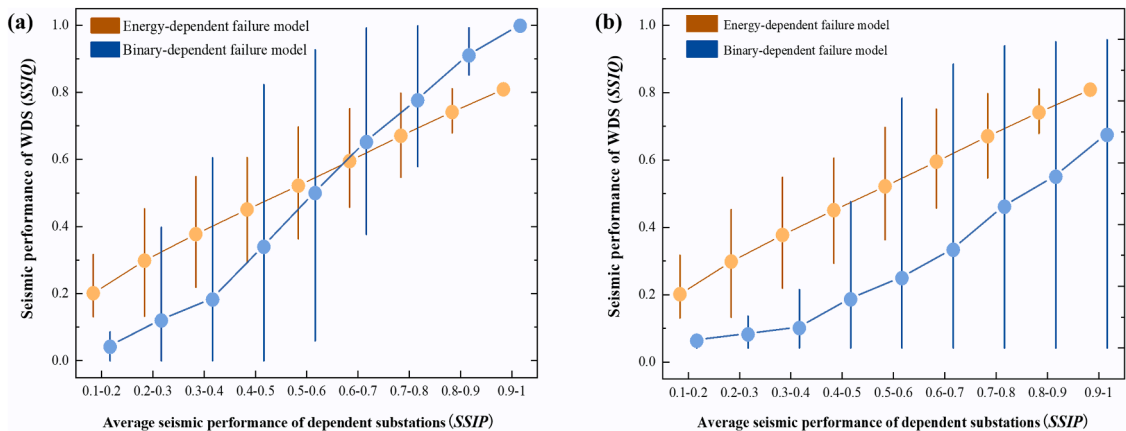


Fig. 10. Seismic performance of Shelby WDS (SSIQ) with different binary-dependent failure thresholds. ($M_W = 7.5$).

(a) Binary-dependent failure model with a threshold of 70%(Logic-70%). (b) Binary-dependent failure model with a threshold of 100%(Logic-100%).

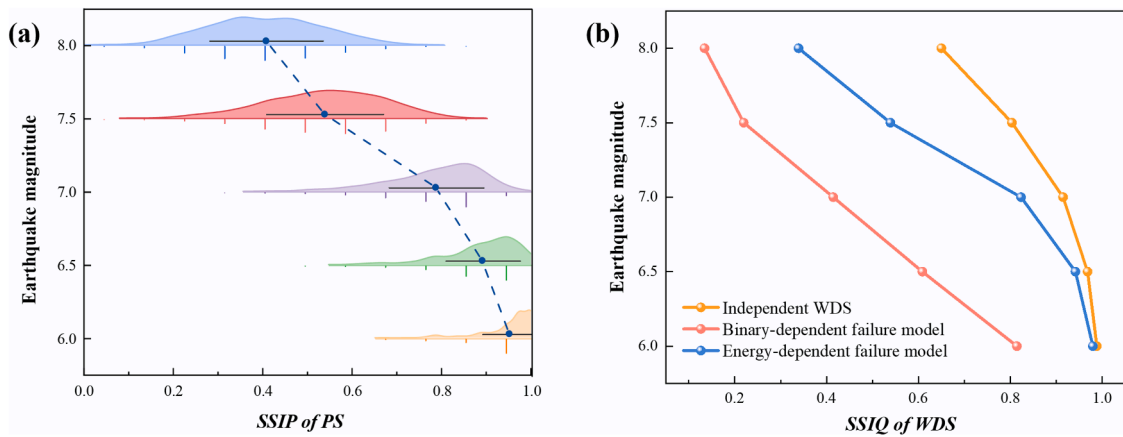


Fig. 11. Seismic performance of the PWS at different earthquake magnitudes in the Shelby case. (a) Seismic performance indicator of the PS. (b) Seismic performance indicator of the WDS with different dependent failure models.

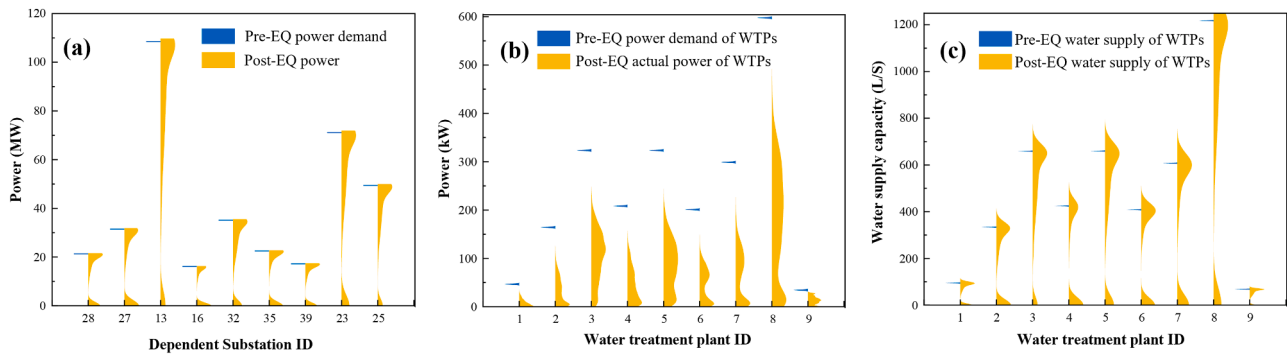


Fig. 12. Supply comparison of dependent substations and water treatment plants before and after the earthquake. (a) Results of the dependent substation power output (post-earthquake to pre-earthquake). (b) Results of the power demand (post-earthquake to pre-earthquake) at water treatment plants (WTPs). (c) Results of the water supply (post-earthquake to pre-earthquake) at the WTPs.

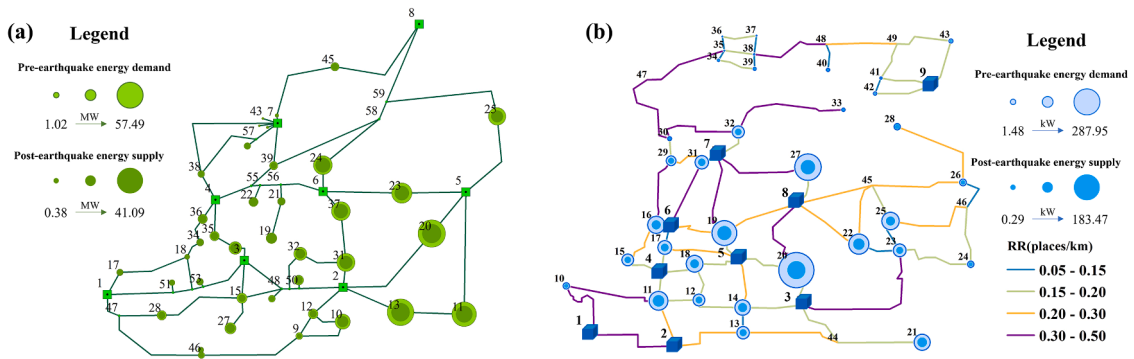


Fig. 13. Spatial diagram of user energy indicators in the PWS. (a) The results of the energy indicators in PS. (b) The results of the energy indicators in WDS.

with varying topologies and functional characteristics, this section selects another PWS case in NY, China. The WDS of this PWS comprises 9 water sources, 72 pipelines, and 49 user nodes, while its PS consists of 4 power plants, 30 110-kV substations, 4 220-kV substations, and 50 distribution lines. The topology of this PWS is presented in Fig. 14.

The analysis results are shown in Fig. 15. Under the identical earthquake scenario, the SSIP of the WDS assessed using the energy-dependent failure model in both the NY and Shelby cases similarly falls between that of the independent operation model and the binary-dependent failure model. Compared with the Shelby case, as earthquake magnitude increases, the discrepancy between the energy-dependent failure model and the independent operation model

increases more quickly, increasing from 16.67% to 86.59%. Meanwhile, the discrepancy between the energy-dependent failure model and the binary dependent failure model shrinks more drastically, decreasing from 83.33% to 13.41%. Two key factors account for this phenomenon: on the one hand, the water treatment plants in NY are more concentrated than those in Shelby County, leading to a higher concentration of WDS nodes dependent on substations; on the other hand, substations in the PS of NY are more prone to regional power outages during earthquakes, which can be observed from the comparison between the Fig. 11 (a) and Fig. 15(a).

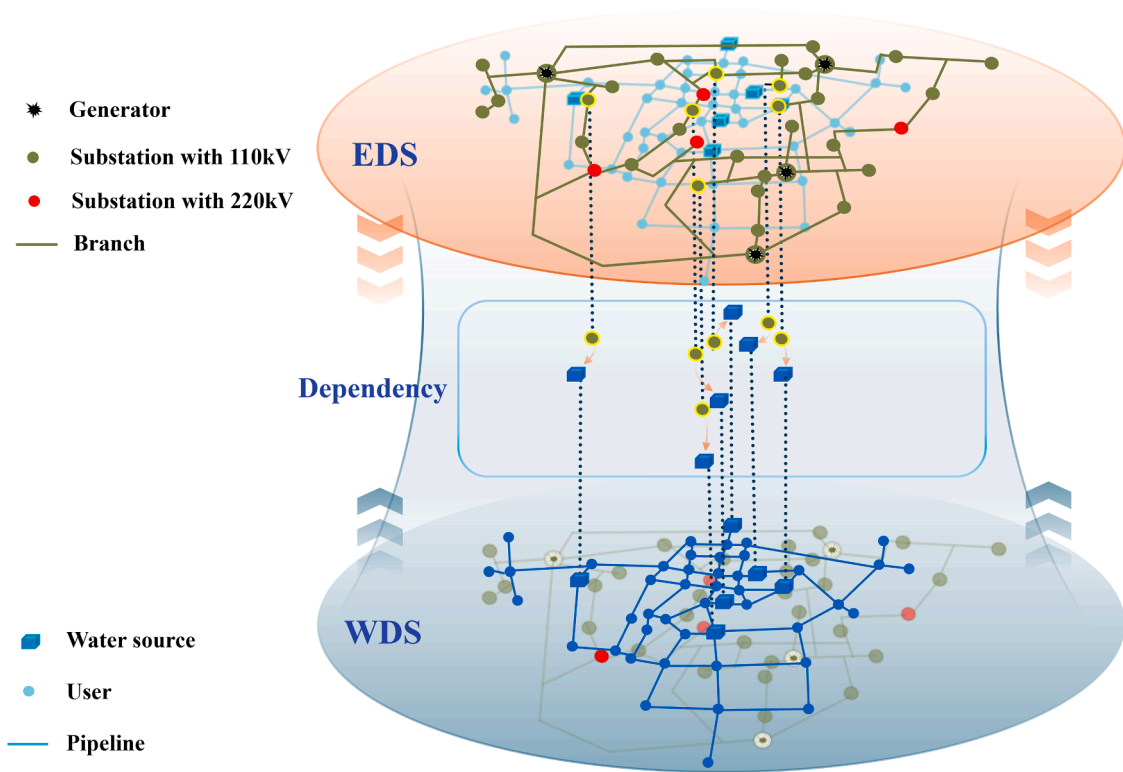


Fig. 14. Schematic diagram of the topology of the PWS in NY.

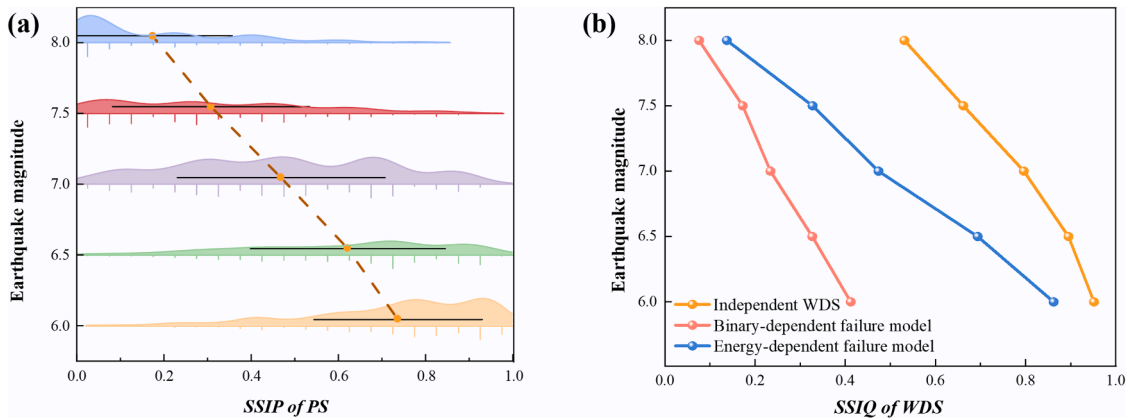


Fig. 15. Seismic performance of the PWS at different earthquake magnitudes in the NY case. (a) Results of the PS. (b) Results of the WDS with different dependent failure models.

3.4. Influence of backup power facility in water pump station

The above analysis indicates that the PS plays a critical role in the normal operation of WDS. Insufficient power supply can severely affect water supply across an entire city. However, not all water treatment plants typically have backup power supplies, and even for those that do, their backup capacity may be insufficient to sustain the normal operation of pumps. While previous analyses focused on scenarios where WTPs completely lacked backup power, this section conducts a parameter sensitivity analysis using the Shelby PWS case. This section conducts the sensitivity analysis to evaluate the impact of backup power supply probability ($P(W|E)$) and the power supply priority for the WTP (α) on the seismic performance of the PWS. Here, α characterizes the priority level of the dependent substation supplying power to the WTP. For instance, $\alpha=0.7$ indicates that as long as the dependent substation's post-earthquake active power remains at or above 70% of its pre-

earthquake capacity, it will prioritize supplying power to the WTP. Consequently, a smaller α value represents a higher emergency power supply priority for the WTP.

Fig. 16(b) shows that the seismic performance of the WDS gradually increases as the $P(W|E)$ increases (green line > blue line > yellow line). For the scenario with $P(W|E)=0.5$, the seismic performance difference between $\alpha=0.3$ and $\alpha=1.0$ is within 6%, corresponding to the maximum and minimum values of the blue line. This suggests that a higher α value (i.e., a lower power supply priority for the WTP during post-disaster power allocation) limits the seismic performance of the WDS. When α is fixed at 0.7 (shown in the subfigure of Fig. 16(b)), the seismic performance of the WDS gradually increases with the rise of the $P(W|E)$, and the seismic performance gradually approaches that of the independent operation model (i.e., the $P(W|E)=100\%$). When the $P(W|E)$ increases from 10% to 90%, the seismic performance of the WDS improves significantly by 29.9%.

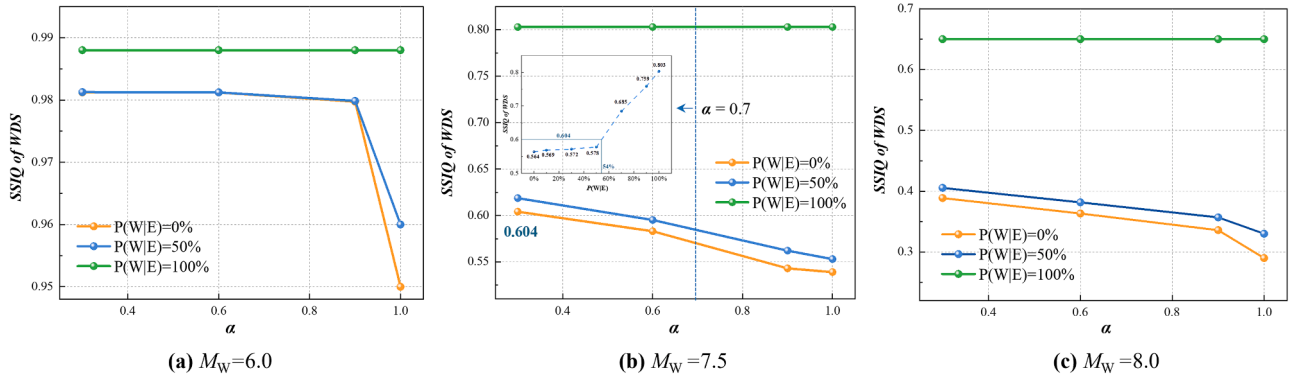


Fig. 16. Sensitivity analysis of the seismic performance model for PWS (Shelby Case). (α : power supply priority for the WTP).

A cross-comparison of Figs. 16(a), 16(b), and 16(c) reveals that under lower earthquake magnitudes, the power supply priority (α) has minimal impact on WDS seismic performance, which is predominantly governed by the backup power availability ($P(W|E)$). However, under higher earthquake magnitude scenarios, both $P(W|E)$ and α exert significant controlling effects on the system performance. Therefore, implementation measures—such as elevating the power supply priority for WTPs (i.e., assigning a smaller α), and ensuring the reliability of backup generators—can effectively boost the seismic performance of the PWS. Specifically, Fig. 16(b) demonstrates that increasing the backup power availability $P(W|E)$ from 0% to 54% yields the same performance

improvement as lowering α from 0.7 to 0.3. Thus, in practical disaster mitigation scenarios where high backup power availability cannot be strictly guaranteed, prioritizing power allocation to WTPs (elevating their supply priority) serves as a highly effective alternative strategy to mitigate the seismic cascading risk of urban PWS.

3.5. Correlation analysis among the different failure models

In the context of city-scale seismic performance assessment, defining a universal power threshold is highly subjective and introduces considerable uncertainty. To demonstrate this, a detailed analysis has

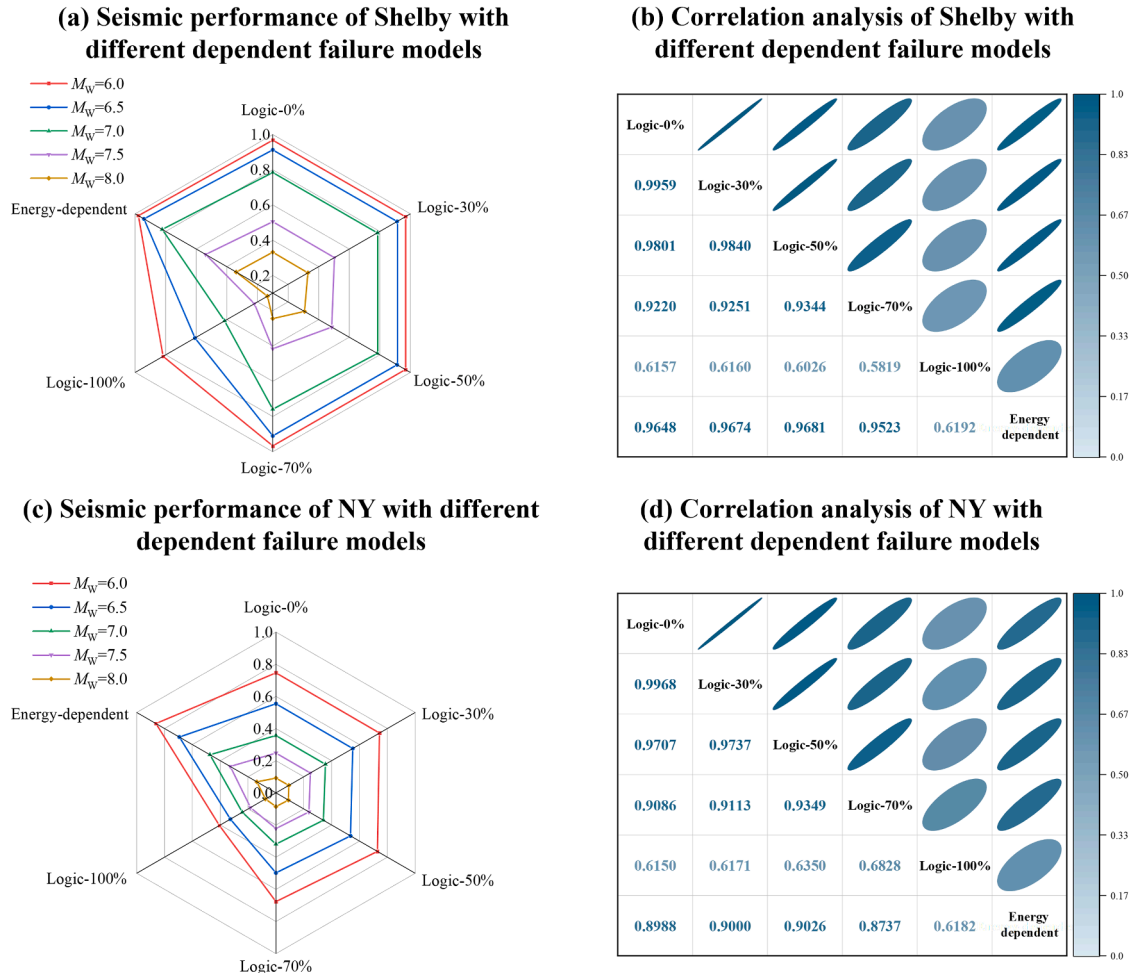


Fig. 17. Comparative analysis of binary-dependent failure models (0%, 30%, 50%, 70%, 100% thresholds) and the energy-dependent failure model.

been conducted to compare various binary threshold scenarios (0%, 30%, 50%, 70%, 100%) against the energy-dependent failure model, using the Shelby and NY as comparative cases (as shown in Fig. 17). Interestingly, the results reveal a critical contradiction inherent in binary-dependent failure models: When evaluating the overall performance degree (as shown in Fig. 17(a) and 17(c)), the binary model with a 0% threshold visually approximates the energy-dependent results best across both cases. However, when assessing the linear behavioral trend (Pearson correlation in Fig. 17(b) and 17(d)), the synchronization shifts significantly—Shelby and NY align most closely with a 50% threshold.

This discrepancy between absolute magnitude and trend correlation highlights a fundamental flaw: a single, fixed threshold in a binary model cannot robustly capture both the overall performance degree and the detail degradation trend in the network. The continuous energy-dependent failure model bypasses this subjective thresholding entirely, providing a much more robust and mathematically stable framework.

4. Conclusions

To investigate the seismic performance of dependent power-water systems (PWS) during earthquake hazards, this study proposes an energy-dependent failure model for the power system (PS) and water distribution system (WDS) based on the power flow models and hydraulic models. This dependent failure model characterizes the cascading failure propagation process between these two heterogeneous infrastructure systems through energy conservation based on the dependent substations and pumps of the water treatment plant. The characteristic analyses on PWS cases are conducted with varying earthquake magnitudes, topologies, and functional characteristics. The main conclusions are summarized as follows:

- (i) The fundamental assumptions of dependency failure models critically dictate the seismic performance outcomes of the PWS, with variations reaching up to 90.9%. The binary-dependent model may overestimate seismic performance losses, and different thresholds led to WDS seismic performance variation ranging from 19.3% to 90.9%. In contrast, the energy-dependent failure model proposed by this study can continuously characterize the impact of partial functional failures in substations on WTP functionality, without relying on threshold parameters.
- (ii) The difference in the seismic performance of PWSs evaluated by the binary-dependent model and the energy-dependent model decreases as the magnitude increases. Accordingly, under rare earthquakes, extensive pipeline damage to the WDS leads to an insignificant difference in the PWS seismic performance evaluation across different dependent failure models.
- (iii) The seismic performance of PWS is highly sensitive to the spatial topology of WTPs. Comparative analysis reveals that systems with highly concentrated WTP layouts (e.g., the NY case) degrade much faster under seismic hazards than decentralized ones. Therefore, the spatial locations of WTPs constitute a critical factor influencing the earthquake risk of the WDS. To reduce the malfunction of WDS caused by the dependent power system failure, the WTP layout planning should be spatially decentralized to prevent concurrent failures due to regional power outages. On the other hand, improving the power supply reliability to the region where these WTPs located is paramount to preventing severe cascading failures.
- (iv) Parameter sensitivity analysis demonstrates that enhancing the availability of backup power supplies at WTPs significantly strengthens the seismic performance of the WDS. Consequently, mandating routine inspections and rigorous maintenance of backup power generators by water management authorities must be considered a vital pre-disaster mitigation measure. Specially, in disaster scenarios where WTP backup power supplies are

unavailable or fail to operate, proactively prioritizing grid power assignment to WTPs (i.e., assign power supply to WTPs by residual power capacity in substations) serves as a highly effective emergency response strategy to maximize the seismic performance of WDS.

Overall, this study concentrates on the seismic performance assessment within the PWS, aiming to reveal the energy transfer amid power system-water system-users with a unidirectional dependency between substations and water treatment plants. However, the PS and the WDS exhibit complex interdependencies, and there are still several limitations in this study:

- (i) The proposed energy-dependent model characterizes the degradation of power transmission from substations to WTPs under seismic hazards through energy conservation among active power, water head, and flow rate. However, the reverse dependency of the PS on the WDS is not addressed. Quantifying the energy dependence of power plants or substation cooling systems on water supply capacity will be a critical focus for future research.
- (ii) This study conducts a steady-state power flow analysis tailored to city-scale distribution networks in evaluating the seismic performance of the PS. However, expanding the spatial scope to a regional scale to incorporate the seismic vulnerability of transmission networks and adopting a detailed power flow model to capture more failure mechanisms present a significant challenge and a crucial direction for future work.
- (iii) Regarding the facility damage to the WDS, this study primarily accounts for pipeline damage and the loss of power supply to WTPs. Incorporating the direct structural damage to WTPs into the proposed framework, and evaluating how such physical damage degrades the overall functionality of the broader PWS, will be another key priority in subsequent studies.

CRedit authorship contribution statement

Tianhe Ma: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation. **Chengshun Xu:** Writing – review & editing, Visualization, Validation, Supervision. **Benwei Hou:** Writing – review & editing, Validation, Supervision, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

- [1] Buldyrev SV, Parshani R, Paul G, Stanley HE, Havlin S. Catastrophic cascade of failures in interdependent networks. *Nature* 2010;464:1025–8. <https://doi.org/10.1038/nature08932>.
- [2] Shen K, Lan M, Gardoni P, Peng W, Xie Z, Geng J, et al. Evaluation of the resilience of large-scale interdependent infrastructure through simplification-enhanced network modeling. *Reliab Eng Syst Saf* 2026;269:112026. <https://doi.org/10.1016/j.res.2025.112026>.
- [3] Hou B, Huang J, Miao H, Zhao X, Wu S. Seismic resilience evaluation of water distribution systems considering hydraulic and water quality performance. *Int J*

- Disaster Risk Reduct 2023;93:103756. <https://doi.org/10.1016/j.ijdrr.2023.103756>.
- [4] Liu W, Xie Z, Xie J. Seismic reliability assessment of in-service water distribution networks. *Probabilistic Eng Mech* 2025;81:103826. <https://doi.org/10.1016/j.probengmech.2025.103826>.
 - [5] van de Lindt JW, Kruse J, Cox DT, Gardoni P, Lee JS, Padgett J, et al. The interdependent networked community resilience modeling environment (IN-CORE). *Resilient Cities Struct* 2023;2:57–66. <https://doi.org/10.1016/j.rcns.2023.07.004>.
 - [6] Reilly AC, Baroud H, Flage R, Gerst MD. Sources of uncertainty in interdependent infrastructure and their implications. *Reliab Eng Syst Saf* 2021;213:107756. <https://doi.org/10.1016/j.res.2021.107756>.
 - [7] Ouyang M. Review on modeling and simulation of interdependent critical infrastructure systems. *Reliab Eng Syst Saf* 2014;121:43–60. <https://doi.org/10.1016/j.res.2013.06.040>.
 - [8] Chang SE, McDaniels TL, Mikawoz J, Peterson K. Infrastructure failure interdependencies in extreme events: power outage consequences in the 1998 Ice Storm. *Nat Hazards* 2007;41:337–58. <https://doi.org/10.1007/s11069-006-9039-4>.
 - [9] Tan RR, Aviso KB, Promentilla MAB, Yu KDS, Santos JR. Fuzzy inoperability input-output analysis of mandatory biodiesel blending programs: the Philippine case. *Energy Procedia* 2014;61:45–8. <https://doi.org/10.1016/j.egypro.2014.11.902>.
 - [10] Kaegi M, Mock R, Kröger W. Analyzing maintenance strategies by agent-based simulations: a feasibility study. *Reliab Eng Syst Saf* 2009;94:1416–21. <https://doi.org/10.1016/j.res.2009.02.002>.
 - [11] Bolwig S, Bazbauers G, Klitkou A, Lund PD, Blumberga A, Gravelins A, et al. Review of modelling energy transitions pathways with application to energy system flexibility. *Renew Sustain Energy Rev* 2019;101:440–52. <https://doi.org/10.1016/j.rser.2018.11.019>.
 - [12] Ouyang M, Wang Z. Resilience assessment of interdependent infrastructure systems: with a focus on joint restoration modeling and analysis. *Reliab Eng Syst Saf* 2015;141:74–82. <https://doi.org/10.1016/j.res.2015.03.011>.
 - [13] Adachi T, Ellingwood BR. Serviceability of earthquake-damaged water systems: effects of electrical power availability and power backup systems on system vulnerability. *Reliab Eng Syst Saf* 2008;93:78–88. <https://doi.org/10.1016/j.res.2006.10.014>.
 - [14] Dueñas-Osorio L, Craig JJ, Goodno BJ. Seismic response of critical interdependent networks. *Earthq Eng Struct Dyn* 2007;36:285–306. <https://doi.org/10.1002/eqe.626>.
 - [15] Anwar GA, Dong Y, Ouyang M. Systems thinking approach to community buildings resilience considering utility networks, interactions, and access to essential facilities. *Bull Earthq Eng* 2023;21:633–61. <https://doi.org/10.1007/s10518-022-01557-y>.
 - [16] Li Y, Zhang M. Cascading failure analysis of interdependent water-power networks based on functional coupling. *Reliab Eng Syst Saf* 2025;259:110950. <https://doi.org/10.1016/j.res.2025.110950>.
 - [17] Ouyang M. A mathematical framework to optimize resilience of interdependent critical infrastructure systems under spatially localized attacks. *Eur J Oper Res* 2017;262:1072–84. <https://doi.org/10.1016/j.ejor.2017.04.022>.
 - [18] Yang S, Zhang Y, Lu X, Guo W, Miao H. Multi-agent deep reinforcement learning based decision support model for resilient community post-hazard recovery. *Reliab Eng Syst Saf* 2024;242:109754. <https://doi.org/10.1016/j.res.2023.109754>.
 - [19] Xiao Y, Zhao X, Wu Y, Chen Z, Gong H, Zhu L, et al. Seismic resilience assessment of urban interdependent lifeline networks. *Reliab Eng Syst Saf* 2022;218:108164. <https://doi.org/10.1016/j.res.2021.108164>.
 - [20] Çağan Z, Davidson RA, Guikema SD. Post-earthquake restoration planning for Los Angeles electric power. *Earthq Spectra* 2006;22:589–608. <https://doi.org/10.1193/1.2222400>.
 - [21] Liu, S. and Zhang, X., Investigation report on earthquake disaster and relief of water supply system based on wenchuan earthquake. Tongji University Press, 2013.(in Chinese).
 - [22] Zhou M, Zhang K, Liu X, Ma F. Data-assisted coordinated robust optimization method for urban water-power systems. *Appl Energy* 2025;388:125670. <https://doi.org/10.1016/j.apenergy.2025.125670>.
 - [23] Ouyang M, Cheng Z, Ma J, Wang H, Mitoulis SA. Coupled urban risks: a complex systems perspective with a people-centric focus. *Engineering* 2025;44:44–50. <https://doi.org/10.1016/j.eng.2024.12.023>.
 - [24] Zhang W, Lin P, Wang N, Nicholson C, Xue X. Probabilistic prediction of postdisaster functionality loss of community building portfolios considering utility disruptions. *J Struct Eng* 2018;144:04018015. [https://doi.org/10.1061/\(ASCE\)ST.1943-541X.0001984](https://doi.org/10.1061/(ASCE)ST.1943-541X.0001984).
 - [25] Wang M, Takada T. Macrospatial correlation model of seismic ground motions. *Earthq Spectra* 2005;21:1137–56. <https://doi.org/10.1193/1.2083887>.
 - [26] Lim H-W, Song J. Efficient risk assessment of lifeline networks under spatially correlated ground motions using selective recursive decomposition algorithm. *Earthq Eng Struct Dyn* 2012;41:1861–82. <https://doi.org/10.1002/eqe.2162>.
 - [27] Zhang L, Hou B, Ma T, Xu C, Miao H, Liu L, et al. Seismic functionality analysis of water distribution network affected by pipeline failure correlations. *Soil Dyn Earthq Eng* 2026;203:110148. <https://doi.org/10.1016/j.soildyn.2026.110148>.
 - [28] Yoon S, Lee Y-J, Jung H-J. Flow-based seismic risk assessment of a water transmission network employing probabilistic seismic hazard analysis. *Nat Hazards* 2021;105:1231–54. <https://doi.org/10.1007/s11069-020-04352-7>.
 - [29] Sharma, N. Regional resilience analysis: modeling, optimization, and uncertainty quantification. Ph.D. University of Illinois at Urbana-Champaign, 2020.
 - [30] Isoyama R, Ishida E, Yune K, Shirozu T. Seismic damage estimation procedure for water supply pipelines. *Water Supply* 2000;18.
 - [31] Liu W, Song Z, Ouyang M, Li J. Recovery-based seismic resilience enhancement strategies of water distribution networks. *Reliab Eng Syst Saf* 2020;203:107088. <https://doi.org/10.1016/j.res.2020.107088>.
 - [32] Hou B, Ma X, Diao K, Zhong Z, Wu S. Seismic performance assessment of water distribution systems based on multi-indexed nodal importance. *Water (Basel)* 2021;13:2362. <https://doi.org/10.3390/w13172362>.
 - [33] Committee, E., Whitman, R., Borchardt, R., Brookshire, D., Eisner, R., Holmes, W., et al. HAZUS-MH 2.1 Earthquake model technical manual. 2013.
 - [34] Shi, P.; O'Rourke, T.D. Seismic response modeling of water supply systems. n.d.
 - [35] Jichao L, Qingxue S, Guanjie H, Quanwang L, Tao W. Functionality analysis of an urban water supply network after strong earthquakes. *Earthq Eng Eng Vib* 2021;20:291–302. <https://doi.org/10.1007/s11803-021-2020-0>.
 - [36] Han Z, Ma D, Hou B, Wang W. Seismic resilience enhancement of urban water distribution system using restoration priority of pipeline damages. *Sustainability* 2020;12:914. <https://doi.org/10.3390/su12030914>.
 - [37] Ma L, Christou V, Bocchini P. Framework for probabilistic simulation of power transmission network performance under hurricanes. *Reliab Eng Syst Saf* 2022;217:108072. <https://doi.org/10.1016/j.res.2021.108072>.
 - [38] Li J, Dueñas-Osorio L, Chen C, Shi C. AC power flow importance measures considering multi-element failures. *Reliab Eng Syst Saf* 2017;160:89–97. <https://doi.org/10.1016/j.res.2016.11.010>.
 - [39] Tiong A, Vergara HA. A two-stage stochastic multi-objective resilience optimization model for network expansion of interdependent power-water networks under disruption. *Int J Crit Infrastruct Prot* 2023;40:100588. <https://doi.org/10.1016/j.ijcip.2023.100588>.
 - [40] Wang, Y., O'Rourke, T.D. Seismic performance evaluation of water supply systems n.d.
 - [41] Jia R, Du K, Song Z, Xu W, Zheng F. Scenario reduction-based simulation method for efficient serviceability assessment of earthquake-damaged water distribution systems. *Reliab Eng Syst Saf* 2024;246:110086. <https://doi.org/10.1016/j.res.2024.110086>.
 - [42] Hou B, Zhang J, Ma T, Miao H, Xu C, Du X. Exploring infrastructure dependence on community environment under natural hazards: quantifying the impact of building destruction on infrastructure functionality. *Int J Disaster Risk Reduct* 2026;133:105935. <https://doi.org/10.1016/j.ijdrr.2025.105935>.
 - [43] Delano JE, Gold RD, Briggs RW, Jibson RW. Coseismic Sackungen in the New Madrid seismic Zone, USA. *Geophys Res Lett* 2018;45. <https://doi.org/10.1029/2018GL080493>. 13,258–13,268.
 - [44] Hou B, Yuan M, Diao K, Ma X, Zhou B, Du X. Probabilistic characteristic analysis of seismic performance of water distribution system based on quasi-Monte Carlo simulation. *Earthq Eng Eng Vib* 2025;24:595–611. <https://doi.org/10.1007/s11803-025-2323-7>.